DERIVED PULL-PUSH FORMULA ALONG AFFINE MORPHISMS: RF_* LF^* EQUALS DERIVED TENSOR WITH F_* OF THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. The source ends 'Some details omitted'; the omitted step is the affine module computation Rf_* Lf^*E = E tensor^L_A B, carried as an explicit obligation.

Editorial note. Both same-chapter inputs (lemma-quasi-coherence-pullback, lemma-affine-compare-bounded) are NOT carded (cross-line-claimed labels) and are declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 915 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Structure sheaf pushforward is derived-acyclic for affine f. Cohomology of Schemes, lemma relative-affine-vanishing (cross-chapter, named). Gives f_*O_X = Rf_*O_X, so E tensor f_*O_X = E tensor Rf_*O_X.
- Quasi-coherence of derived pullback. lemma-quasi-coherence-pullback (same chapter, NOT carded, declared external). Places Lf^*E in D_QCoh(O_X).
- Affine comparison descriptions. lemma-affine-compare-bounded (same chapter, NOT carded, declared external). The affine-case identification of D_QCoh with module complexes.

Source reference. Original reference 01XC: focused-seeds/geometry/S9-perf2-affine-morphism-pull-push/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08DW: sources/stacks-perfect/pinned-original-source.tex, source lines 581–581. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be an affine morphism of schemes. For E in $D_{QCoh}(\mathcal{O}_S)$ we have $Rf_*Lf^*E = E \otimes^{\mathbf{L}}_{\mathcal{O}_S} f_*\mathcal{O}_X$.*

Proof. Since f is affine the map $f_*\mathcal{O}_X \to Rf_*\mathcal{O}_X$ is an isomorphism (Cohomology of Schemes, Lemma 01XC). There is a canonical map $E \otimes^{\mathbf{L}} f_*\mathcal{O}_X = E \otimes^{\mathbf{L}} Rf_*\mathcal{O}_X \to Rf_*Lf^*E$ adjoint to the map

$$Lf^*(E \otimes^{\mathbf{L}} Rf_*\mathcal{O}_X) = Lf^*E \otimes^{\mathbf{L}} Lf^*Rf_*\mathcal{O}_X \longrightarrow Lf^*E \otimes^{\mathbf{L}} \mathcal{O}_X = Lf^*E$$

coming from $1 : Lf^*E \to Lf^*E$ and the canonical map $Lf^*Rf_*\mathcal{O}_X \to \mathcal{O}_X$. To check the map so constructed is an isomorphism we may work locally on S. Hence we may assume S and therefore X is affine. In this case the statement is clear from the description of the derived categories $D_{QCoh}(\mathcal{O}_X)$ and $D_{QCoh}(\mathcal{O}_S)$ and the functor Lf^* given in Lemmas 06Z0 and 08DW. Some details omitted. □